

A-FLOWS

Austrian Flood Loss Observation and Web Service

Final project report

Submitted by

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1. Project Overview

The A-FLOWS project has been developed as part of the IP:SDI course of the Applied Geoinformatics Master's program in the winter term 2025/2026. First chapter contains overview of the project, which includes general information, the project's practical context, information about stakeholders, target groups, goals and objectives within the scope of the project.

1.1 General Information

Project Name	A-FLOWS Austrian Flood Loss Observation and Web Service
Project Acronym	A-FLOWS
Project Members	Md Saleh Shakeel Nomaan Denis Vasin Qinwei Zhu
Project Deliverable	Dashboard, Code and Paper/Documentation
Start Date	09/11/2025
End Date	31/01/2026

1.2 Context

In September 2024, Austria was struck by one of its most severe flood disasters in recent years, resulting in damages estimated at €1.3 billion and revealing critical gaps in household-level protection and state emergency response. Examining the problems of flooding in Austria it is clear that the number of flood events has been on the rise in both towns and the countryside, a trend that will only become more serious as time goes by. Floods pose a threat to lives, homes, communities and the very infrastructure that we rely on, and climate change and human activities just exacerbate these risks, that is why desperate measures are needed for high-resolution systems that will help us to understand the risks, plan for emergencies and make smart decisions.

It is the EU's Floods Directive that drives the systematic assessment and mapping of flood risks. It lays down the need for the systematic assessment, mapping and governance of flood risks and Austrian's official flood hazard maps and flood zones, or 'return periods' are used to pinpoint areas that are most likely to be hit by floods.

To bring together all the data that is required for reasonable response, Spatial Data Infrastructure, or SDI should be used.

1.3 Concept and Objectives

This project aims to build up a tool for bringing together different sources of geographical data, with the mixture of hazard maps and population statistics allowing all parties involved to see exactly where people live in relation to the area's flood zones. Within this context, the A-FLOWS project focuses on demonstrating of how an open, standardized SDI-based approach can bring together official maps of flood hazards and demographic information, and its key

feature is visualization and time-saving analysis, making it so all affected stakeholders at the time of need could properly react to the outcomes of disaster. This goal will be reached by building an interactive online dashboard. Sub-goals include:

- Analysis: Analysing demographic and vulnerability factors affecting populations at risk.
- SDI Architecture: Organize official HQn and Zählsprenkel data in an ISO 13249-3 compliant PostGIS Enterprise Geodatabase.
- Standardization: Document all datasets using the ISO 19115 metadata standard and publish resources via OGC View (WMS) and Download (WFS/WCS) Services.
- Visualization: Implement a dashboard to visualize the standardized OGC services, providing actionable insights for emergency management.

1.4 Non-Goals

A-FLOWS will not develop the real-time forecasting of flood events. The project does not include the acquisition of new survey data from households.

1.5 Stakeholders

For the A-FLOWS project in Austria, the stakeholders that are counting on its results are public bodies, emergency management services and town planners who specialize in flood control and urban planning. Well-known at risk are the elderly above 65 years old and children up to 15 years old who can really struggle in the aftermath of a flood, often because they are not as mobile and need ongoing healthcare or rely on others for help. Because these groups are so important in the event of a flood, the project has a particular focus on them.

Local authorities in areas prone to flooding are also key stakeholders in the project, since they get a lot out of having access to super-accurate, location-based information that can help them educate and prepare their communities for floods.

1.6 Methodology

We have utilized a Spatial Data Infrastructure (SDI) framework to combine official flood hazard data, demographic statistics, and shelter locations, when building the A-FLOWS project. Our approach to this task was governed by standardized geospatial best practices. We collected, pre-processed, and harmonized three flood hazard datasets (HQ30, HQ100, and HQ300) and demographic data at the Zählsprenkel level, converting them to a unified coordinate reference system, the Austrian national CRS, MGI/Austria Lambert, EPSG:31287.

The prepared data was logically grouped by theme and made to work together, using spatial combination techniques to figure out how many people would be exposed to different flood scenarios, taking a particular interest in the welfare of the elderly and children. Our datasets are documented according to ISO 19115 metadata standards, and are prepared for publication through WMS and WFS services, which will allow for real-time visualization and communication of the results through interactive maps and dashboards.

2. Work Plan

The work plan presented on the next picture (Figure 1) consists of five work packages which include: overall project management (WP1), literature review and collection of data (WP2, WP3) and final packages of creation of methodology and final discussion of a project (WP4, WP5).

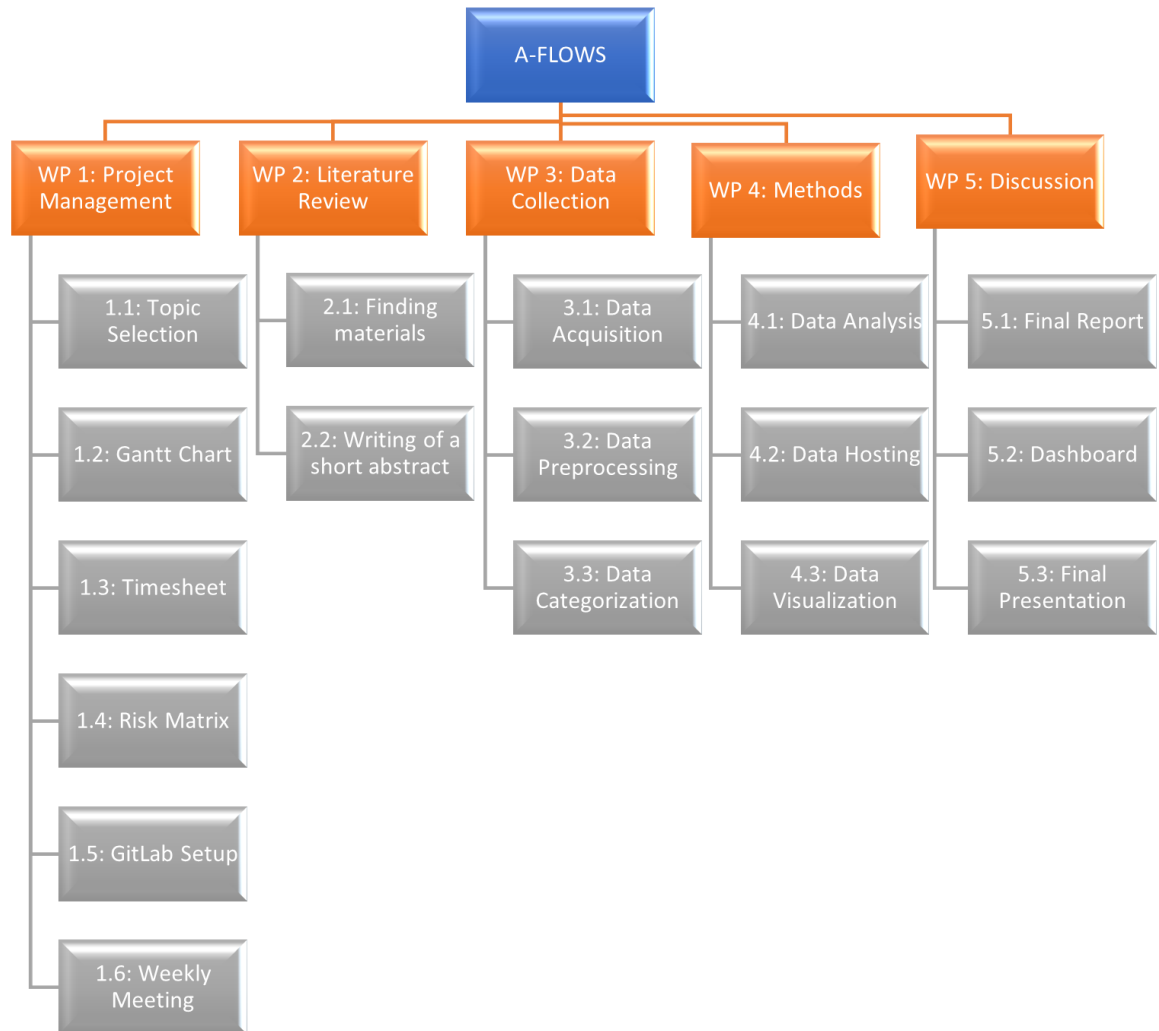


Figure 1: Work plan break down structure

2.1 WP1: Project Management

Objectives and Tasks

Effective project management is key to a well-structured and cohesive operation. This package is responsible for the coordination, documentation and organisation of the project's activities and its attention is focused on getting the team working together, hitting important deadlines, and making sure that everyone is on the same page, and recording the progress of the project in a completely transparent way. This all provides a foundation that helps the project stay on track in accordance with the requirements of the IP:SDI course. For A-FLOWS, project management was mainly conducted through the GitLab.

Gantt chart

The Gantt chart (Table 1) is a visualization and scheduling tool for project management. It helped to mark out all WPs and tasks of the GEPAUS project as well as following which member of a team is doing what during the work on the project

Table 1. GanttChart of A-FLOWS project

WP	Task	Start Date	End Date	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	Status	Progress
				28-03Nov	04-10Nov	11-17Nov	18-24Nov	25-Nov-01Dec	02-08Dec	09-15Dec	16-22Dec	23-29Dec	30Dec-05Jan	06-12Jan	13-19Jan		
1 Project Management																	Ongoing
	1.1 Topic Selection	28-Oct	04-Nov														Completed
	1.2 Gantt Chart	04-Nov	20-Jan														Completed
	1.3 Timesheets	04-Nov	20-Jan														Completed
	1.4 Risk Matrix	04-Nov	01-Dec														Completed
	1.5 GitLab Setup	04-Nov	01-Dec														Completed
	1.6 Weekly Meetup	09-Nov	20-Jan														Completed
2 Literature Review																	Ongoing
	2.1 Finding Materials	09-Nov	01-Dec														Completed
	2.2 Short Abstract	11-Nov	01-Dec														Completed
3 Data Collection																	Ongoing
	3.1 Data Acquisition	01-Dec	07-Dec														Completed
	3.2 Data Pre-Processing	07-Dec	14-Dec														Completed
	3.3 Data Categorization	14-Dec	20-Dec														Completed
4 Methods																	Ongoing
	4.1 Data Analysis	20-Dec	20-Jan														Completed
	4.2 Data Hosting	20-Jan	22-Jan														Completed
	4.3 Data Visualization	22-Jan	27-Jan														Completed
5 Discussion																	Ongoing
	5.1 Final Report	29-Dec	27-Jan														Completed
	5.2 Final Presentation	25-Jan	31-Jan														Completed

GitLab

The GitLab of the A-FLOWS project features a Wiki page that gives an overall view on the project and issues that were divided into Working Packages and regularly updated, which helped with keeping track of an open tasks. The repository of the GitLab contains shared data, documents and metadata.

2.2 WP2: Literature Review

Objectives and Tasks

The objective of this work package was in reviewing the scientific and technical literature to establish a theoretical basis for the project, when building the A-FLOWS project. The literature review concentrated on the best approaches in the design and deployment of Spatial Data Infrastructures (SDI), as well as recent implementations of ISO standards for geospatial data and metadata, and the geospatial analytical methods that have been used to assess flood exposure.

Since flood exposure is closely tied to demographic dynamics, the work gave particular attention to techniques that merge flood hazard with population data, and explored area-based overlay methods including the area proportional method.

Workflow

After an initial discussion of all major aspects of the project within the team, a short abstract was formulated to clearly define and consolidate the project topic. This step was needed to ensure a shared understanding of the project scope and to prevent significant deviations from the chosen topic during later project phases. This topic was basically chosen for its social importance, escalating flood impacts, and the fact that the Austrian government has made official hazard and demographic data easily available to us.

The next step would be to concentrate on acquiring data and discussing the methods we would use in a future analysis.

Data Acquisition

Our area of interest is the entire country of Austria, and we have picked this area so that we can mash up national flood hazard maps with the local demographics at the Zählsprenkel level. It also allows us to tap into official national datasets and conduct analyses that are relevant to flood management and planning, as well as supporting evacuation and public safety efforts across the country. Flood hazard data for different levels of flooding: HQ30, HQ100 and HQ300, were snatched from the EU Floods Directive, and demographic data at the Zählsprenkel level were obtained from official statistical sources and contain information about how many people live in each area, what their make-up is, and what sort of households they have. We also found the locations of emergency shelters, and made sure that all of these datasets were complete, accurately referenced in space, and suitable for integration into our SDI-based workflow.

The method we used, when conducting the exposure analysis in this project in the absence of finer-scale population distribution data is an area-proportional method that have become a popular solution to calculate population exposure, and used a lot in a large-scale flood risk assessments. These methods, while often an oversimplification of the actual population distribution, tend to give results that are usually appropriate and reliable. Disaster and flood studies have established that the impact of hazard events is more devastating to communities if accessibility is not considered.

2.3 WP3: Data Collection

Data Acquisition and Temporal Characteristics

We've woven together longitudinal demographic statistics and extremely detailed hydrological hazard models, when evaluating the empirical basis of this research. Our demographic data, courtesy of Statistik Austria, is a hybrid combination of two separate data sets, each focusing on different aspects of risk. Our diachronic dataset covering 2002 through the 2025 projection has primarily been used to track the exposure trends and population dynamics over a 23-year period, and is essentially the backbone of this study. Our high-dimensional synchronic dataset, however, is a snapshot from the 2025 horizon that includes the precise socio-structural factors needed to measure the vulnerability. The demographic snapshot for the 2025 horizon isolates the distribution of men and women, age-specific dependency ratios and household types, with single-person households being an important indicator of social isolation and evacuation capability. We cross-referenced the population characteristics with Austria's official flood hazard datasets (HORA/Riskogeb), comprising vector-based flood zones that return period polygons for HQ30 and HQ100 and the intensity-based HazardArea raster, which gives us the means to precisely pin down where and to what extent people are at risk of floods.

Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Wien).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Vorarlberg).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Tirol).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Steiermark).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Salzburg).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Oberösterreich).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Burgenland).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Niederösterreich).csv
Bevölkerung zu Jahresbeginn nach Alter Geschlecht sowie österreichischer ausländischer Staatsangehörigkeit seit 01.01.2002 (Kärnten).csv

Figure 2. Sub-tables for each state extracted from the master table

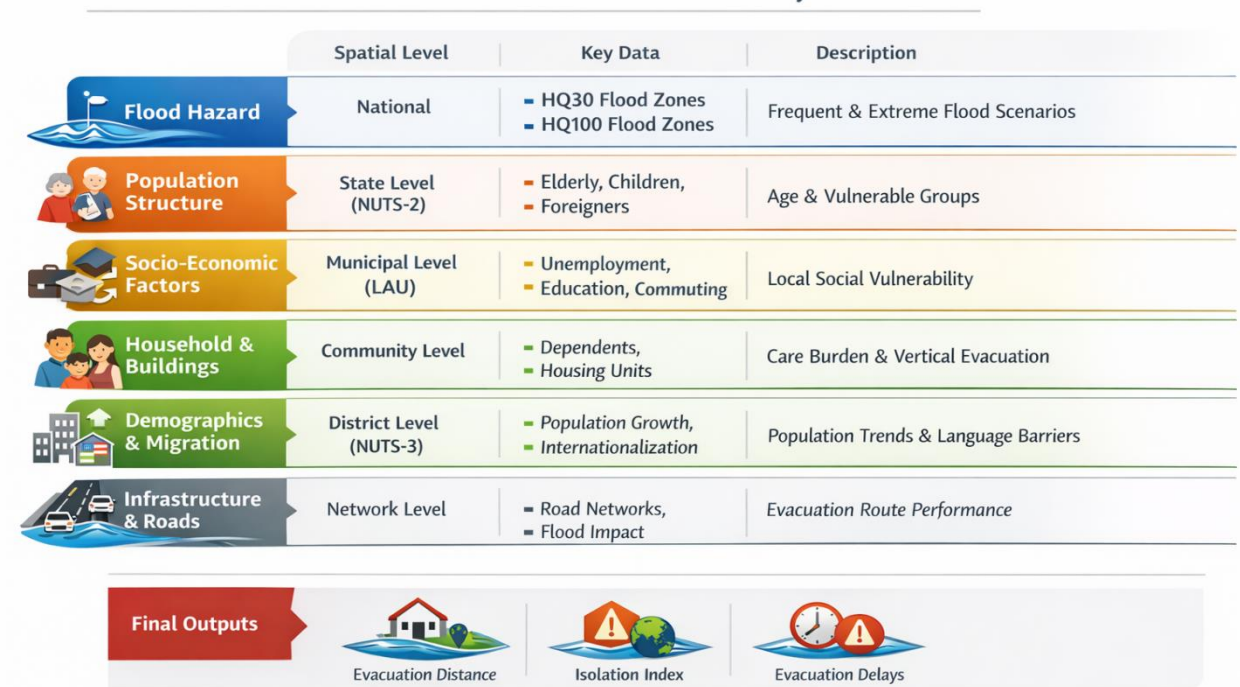


Figure 3. Data for the analysis

Structural Decomposition and Multi-scalar Disaggregation

When tackling the problem of dealing with the raw demographic files in this work, one of the main hurdles was the structural heterogeneity of the data. It was presented in a huge monolithic repository that combined lots of different levels of administrative granularity into a single file. Well-known techniques won't cut it here, so a hierarchical decomposition was developed to take the data apart into three very specific levels: NUTS-2 Federal Provinces, NUTS-3 Districts, and Zählsprengel Census Tracts. We needed this level of detail to make sure that every statistical feature we calculated is accurately linked to the area it represents, otherwise we'd risk counting the same data twice or losing the finer points of the data. Coming down to the Zählsprengel level, about 8,800 tiny units, allowed us to zero in on the high-resolution characteristics of the 2025 household data, that get watered down in higher level groupings. We ran a “Sum-to-Parent” test to double-check that the micro-counts in each unit match up perfectly with the larger administrative totals.

Encoding Normalization and Cross-lingual Harmonization

A tricky pre-processing stage was necessary to iron out differences in character encoding and language terminology, when merging the Austrian census data with the spatial vector layers. Coming from the statistical records of Statistik Austria, the data were in ISO-8859-1, which isn't great at handling German characters, like Umlauts (ä, ö, ü) and the Eszett (ß), that can get mangled when fed into modern GIS systems. Well-known toponyms, which need to be accurately distinguished, have been converted to a standard UTF-8 format to preserve their meaning, and we also implemented a clever cross-lingual harmonization strategy to reconcile the German-language statistical tables with the English-labeled vector files. For the micro-level Zählsprengel and NUTS-3 layers, we used standardized numerical IDs called Lokal-Codes, but the NUTS-2 level needed a bilingual mapping dictionary that could translate between the German administrative names and their English equivalents. This dictionary, which is based on string normalisation and fuzzy matching, managed to secure a 100% join rate for the 2025 structural data.

```
df = pd.read_csv(f_path, encoding='iso-8859-1', dtype={0: str})

df['g_id'] = df.iloc[:, 0].astype(str).str.strip().str.zfill(5)
```

Figure 4. Sub-tables for each state extracted from the master table

Spatial Database Integration and Analysis-Ready Datasets

The cleaned demographic and hydrological data were merged into a unified PostgreSQL/PostGIS spatial database in the Austrian Lambert Projection (EPSG:31287), when it came to the final stage of WP3. Since the demographic units and the flood hazard zones were both in vector format, the integration of the two data types used advanced Polygon-on-Polygon overlay techniques. The Exposed Population Share was calculated by cutting through the flood polygons and the census tracts at the same time. The HazardArea components, being based on raster data, used zone-based statistical techniques to work out the average level of hazard for each and every census district, essentially turning hard numbers into a social and spatial sense of risk. Well-known Analysis-Ready Datasets (ARD) were the result of this process, where we've combined a 23-year population growth rate with our highly detailed 2025 vulnerability profiles.

2.4 WP4: Methodology of work with Data

Methodological Objectives and Strategic Design

This Work Package focused on developing a sound analytical framework to quantify the impact of flood risk to selected vulnerable demographics. Moving from pure descriptive statistics, this phase aimed at creating an added value layer of spatial information by coupling physical hazard intensities with socioeconomic sensitivities. A key aspect of this WP consisted in guaranteeing interoperability of the results, and therefore the methodological chain was developed to deliver standardized spatial outputs compatible with OGC-compliant services (WMS/WFS) and web-based GIS clients, allowing an easy integration within the A-FLOWS decision-support platform.

In our research framework, we define evacuation risk as a multifaceted phenomenon that transcends simple hydrological events, manifesting instead as a complex interplay between environmental hazards, the social fabric of the community, and the performance of critical infrastructure. By implementing a cascading analytical logic, we transition from the physical dimension of flood exposure into layered assessments of social vulnerability and household structures, ultimately intersecting these findings with road network performance to derive our composite risk indices. This structured approach allows us to quantify precisely how physical threats are filtered and amplified by a community's specific demographic and structural characteristics, addressing the common misconception in disaster management that all residents possess equal resilience. To operationalize this, we evaluate the social vulnerability system across two distinct scales: at the macro or state level, we focus on population sensitivity through the weighted proportions of the elderly, children, and foreign nationals, while at the local municipal level, we measure social capacity using unemployment ratios and educational attainment as proxies for resource access. This social data is integrated with our analysis of household dynamics, specifically through the HH_Vuln index which calculates the care burden by assessing the ratio of dependents to working-age residents within individual units, and the Vert_Exp index which distinguishes between those who can seek vertical shelter and those forced into horizontal evacuation. The reliability of these self-derived indicators was ensured through a rigorous experimental process where I personally performed data acquisition from Statistik Austria and managed a comprehensive cleaning phase that involved secondary verification of administrative IDs across various spatial levels, including NUTS-2, LAU, and NUTS-3. This was necessary because an initial attempt revealed boundary mismatches between the statistical census data and the vector shapefiles, leading me to re-execute the entire modeling process to guarantee a precise 1:1 spatial join. Ultimately, by mapping these meticulously cleaned and proprietary indices onto the physical constraints of the road network, we produce final metrics like Socially-Weighted Isolation and Evacuation Time Delay, which reveal that social disadvantages act as a force multiplier that amplifies physical distance and creates high-risk hotspots where residents are both physically and socially stranded.

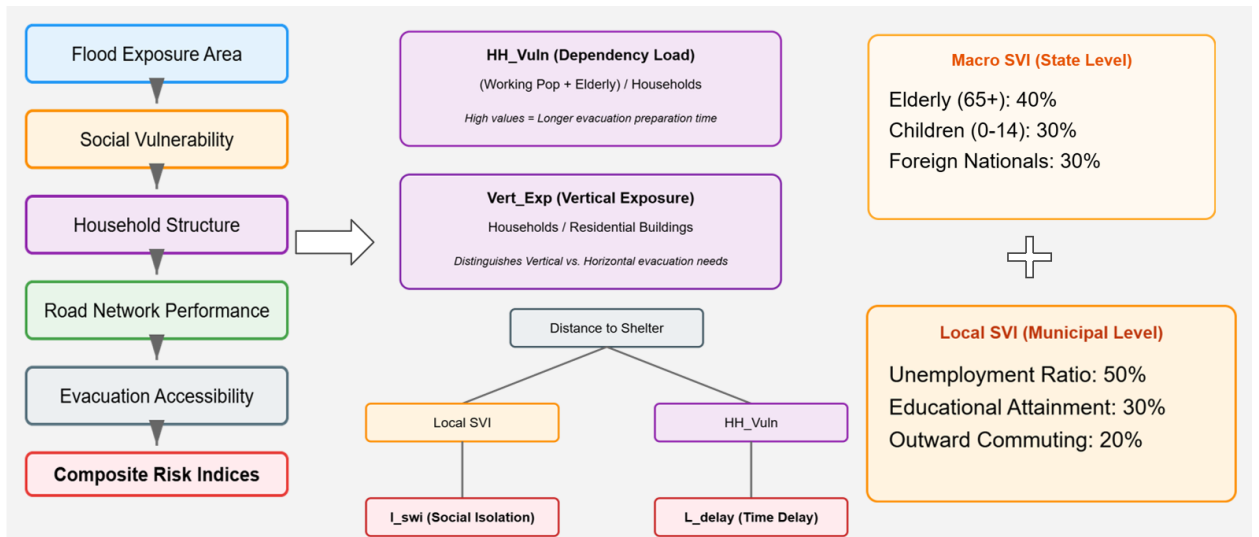


Figure 5. Analysis structure

Workflow

Social Vulnerability Indexing (SVI)

The creation of the Social Vulnerability Index (SVI) acts as the main tool with which to examine the concept of ‘disaster impact inequality.’ In order to narrow down the space between physical distance and deprivation, three different models of calculation have been put in place:

Macro Social Vulnerability Index (Macro SVI)

In order to estimate the baseline demand for rescue at the state level, the Macro Social Vulnerability Index (Macro SVI) results from a weighted combination of demographic sensitivities; the scoring system reflects the dependency ratios of the elderly population and the children, respectively:

$$SVI_{\text{Score}} = (\text{OLD_PCT} * 0.4) + (\text{KIDS_PCT} * 0.3) + (\text{FOR_PCT} * 0.3)$$

The elderly in the population (having a 0.4 weight value) represent a vital focal point, owing to their probable mobility impairments and magnitude of medical dependencies in natural calamities such as floods, resulting in a narrow "evacuation window."

Socially Weighted Isolation Index (I_{swi})

Another misconception activity is in disaster management, where people often believe that distance to shelter means safety. We addressed this misconception by using I_{swi} , where it geometrically increases the Euclidean distance to the shelter, HubDist, by a multiple of the local vulnerability factor:

$$I_{\text{swi}} = \text{HubDist} * (1 + \text{Local_SVI})$$

Exposure Analysis

The approach taken is a regionalized area-proportional method that estimates the number of people exposed to the threat of flooding, when assessing the exposure of vulnerable population groups to flood hazards.

$$\text{Exposure}_{\text{Vulnerable, HQn}} = \sum_{i=1}^N \left(\text{Pop}_{\text{Vulnerable},i} \times \frac{\text{Area}(\text{HQn} \cap \text{Zählsprengel}_i)}{\text{Area}(\text{Zählsprengel}_i)} \right)$$

- $\text{Pop}_{\text{Vulnerable},i}$: The actual count of a **specific vulnerable group** (e.g., population aged 65 and over, or number of single-parent households) in the i -th Zählsprengel (derived from your ODS tables).
- $\text{Area}(\text{HQn} \cap \text{Zählsprengel}_i)$: The **overlapping area** between the HQn flood zone and Zählsprengel i .

Figure 6: Area proportional method

Taking each Zählsprengel, or statistical areas, into account, this approach weighs the size of the elderly population and children living in these areas by the proportion of the area that overlaps with given flood hazard zones, such as HQ30 or HQ100, for these specific groups.

Since it is not possible to perfectly map population distribution and does not account for the potential non-uniform distribution within each Zählsprengel it uses the assumption that the population is uniformly distributed, and offers a reasonable approximation. This gives an average number of people that could be affected by a flood event, and by summing the weighted values for all Zählsprengel the total number of exposed people under different types of floods is made clear.

In order to determine the exposure of susceptible population at risk within the HQ30 and HQ100 hazard zones, the study proposed a regionalized area-proportional approach. The availability of demographic statistics aggregated at the level of the Zählsprengel statistical enumeration district makes it possible to estimate population at risk based on the intersection of residence polygons and flood hazard zones.

Although this approach presumes a smooth distribution of population within each individual Zählsprengel, such an approximation is necessary by default, owing to the absence of relevant dasymetric data at such minute scales. However, it is worth saying that this approach generates a statistically accurate level of exposure. As far as this calculation is concerned, within ArcGIS Pro, this calculation was facilitated through Intersect and Summarize Within.

Accessibility Modeling and Cost Function Design

In evaluating the capacities of the evacuees, a shortest path routing analysis was used in determining the accessible emergency shelters for the exposed populations. Using the ArcGIS Pro Network Analyst extension tool, a dynamic cost function was utilized. Refuting the ideal of "all-weather-network-connectivity," the model locates road segments intersecting with flood zones (). The impedance of the network was adjusted in accordance with real-world factors, slowing pedestrian speeds from 5 km/h to 0.5 km/h in flood zones. Utilizing the Evacuation Delay Index, areas in which displaced populations could possibly miss the optimal time window due to high internal household complexities are identified:

$$L_delay = HubDist * HH_Vuln.$$

The "calculation formula" here is the **path cost function**, which acts as an **implicit weighting**:

$$Cost = Distance_{Segment} / Speed_{Limit}$$

- **Distance_{Segment}**: The physical length of each road segment.
- **Speed_{Limit}**: The **velocity weight** assigned to the road type (e.g., pedestrian speed of 5 km/h).
- **Final Output**: A map layer showing the **time in minutes** from the Zählsprenzel centroid to the nearest Shelter.

Figure 7: Cost function and Weighting

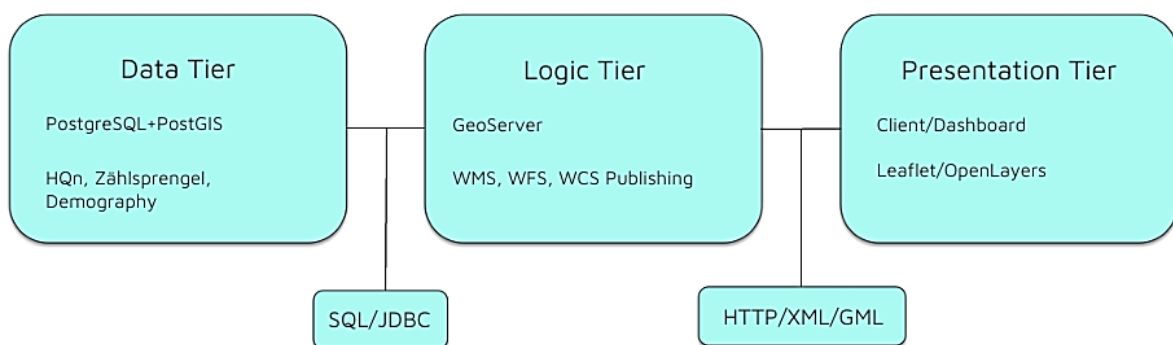
The analysis also takes into account the distance of our target groups to emergency shelters, calculating the shortest path to the nearest shelter for each of the Zählsprenzel centroids, when evaluating the accessibility of the population exposed to floods. The road network used in this routing is defined by a cost function that is basically a measure of how long it takes to travel along each road segment. With the cost of each segment being determined by the physical length of the road and an estimated walking speed.

Well-known shortest-path routing techniques and calculations are applied to find the minimum travel time from the centroids within the flood hazard zones to the nearest shelters. The results give the route geometries and estimated times of up to two minutes, which will be used as indicators of how accessible the areas are to evacuation.

Data Standardization and SDI Integration

For the outputs to be functional spatial assets, all derived indicators (e.g.,) were prepared to be compatible with PostGIS, with ISO 19115-compliant metadata encapsulating all results. To achieve this, by using these results within GeoServer as OGC-compliant services, the methodology ensures that there is reproducibility to the analytical logic, thereby allowing it to be accessible in real-time emergency resource allocation through the use of the A-FLOWS web service.

Our results are stored in a centralized geodatabase that keeps the data neatly organized. This geodatabase standardizes the storage of analysis outputs and allows for the re-use of data across the board. We have also made sure that our hosted data is ready for publication, with interoperable geospatial services and can easily be dropped into a wider SDI framework.



• *Figure 8: SDI Architecture*

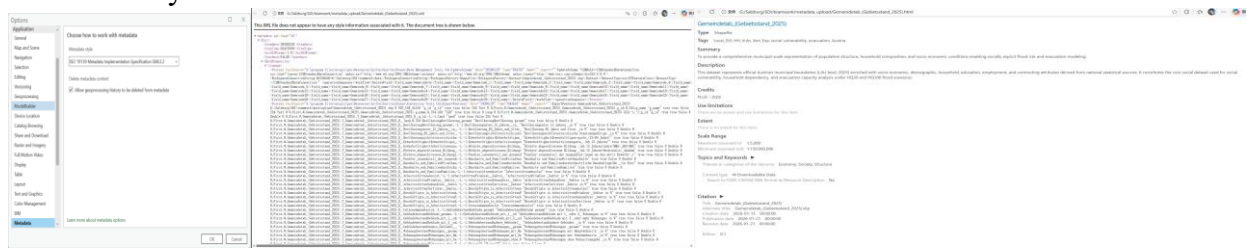
Services

In A-FLOWS, geospatial services are put in place so that analytical outcomes can be got at in a uniform way, work together, and be used again on various platforms – and by different people using them. All the datasets that are worked on, like areas at risk of flooding (HQ30/HQ100), population figures for Zählsprengel areas, social weakness indexes, and how easy it is to get people out, are kept in a central PostgreSQL/PostGIS geodatabase: this is the main place the data is stored. These datasets are then made public via GeoServer as WMS – Web Map Services – that follow OGC standards, for looking at, and WFS – Web Feature Services – for getting the data and doing more work with it; this lets them fit smoothly into GIS clients and web apps. The architecture based on services means the results of analysis can be taken in by the ArcGIS Experience Builder dashboard as they’re created.

Metadata

To guarantee the highest level of transparency, reproducibility, and cross-platform interoperability of the spatial datasets, comprehensive metadata has been scrupulously compiled for all generated shapefiles, including critical layers such as and . The documentation process followed the international geographic metadata standard ISO 19139 strictly and allowed seamless integration into broader SDI environments. Each metadata record encompasses the basic bibliographic and technical descriptors, including thematic keywords, production timelines, and precise definitions of attribute fields. Particular emphasis was placed on defining the analytical significance and operational thresholds of derived indicators, such as the Social Weighted Isolation Index . It was by setting up this rigorous metadata framework that the study made sure that any future researcher or emergency management stakeholder will be able to

interpret the sociodemographic weighting logic and the hydrological constraints embedded in the spatial models correctly and will enhance the longstanding utility of the integrated flood evacuation system.

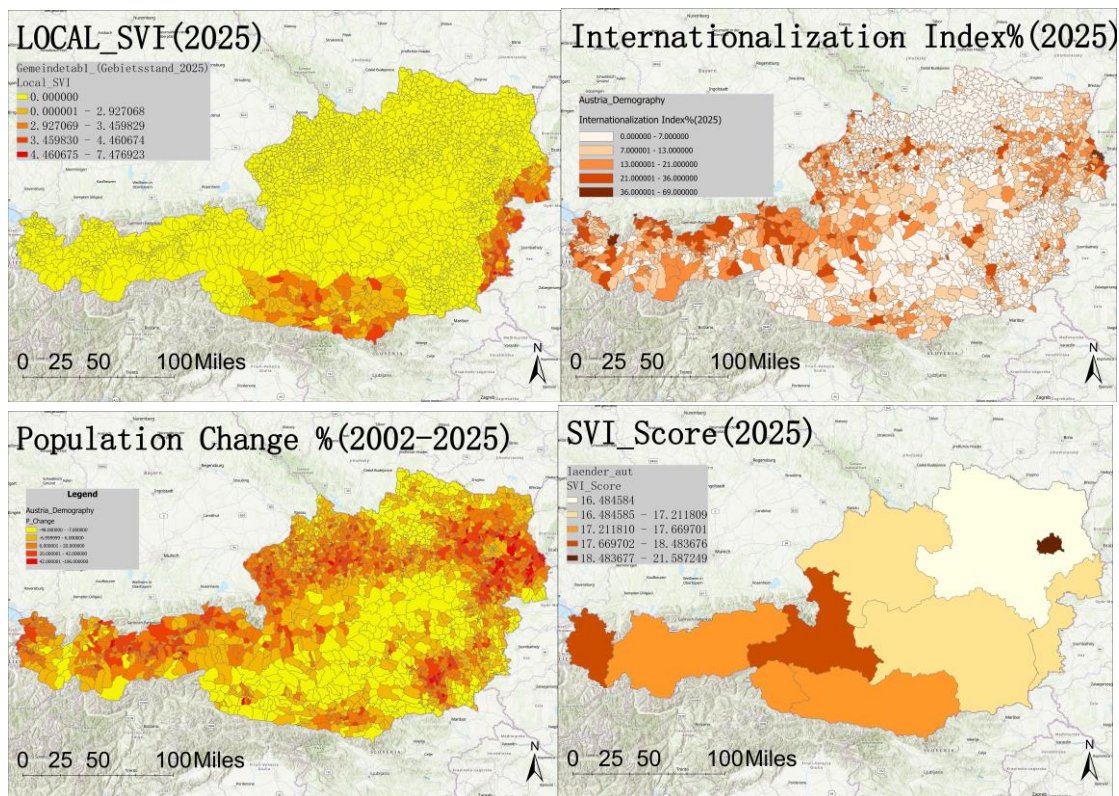


- *Figure 9:Metadata*

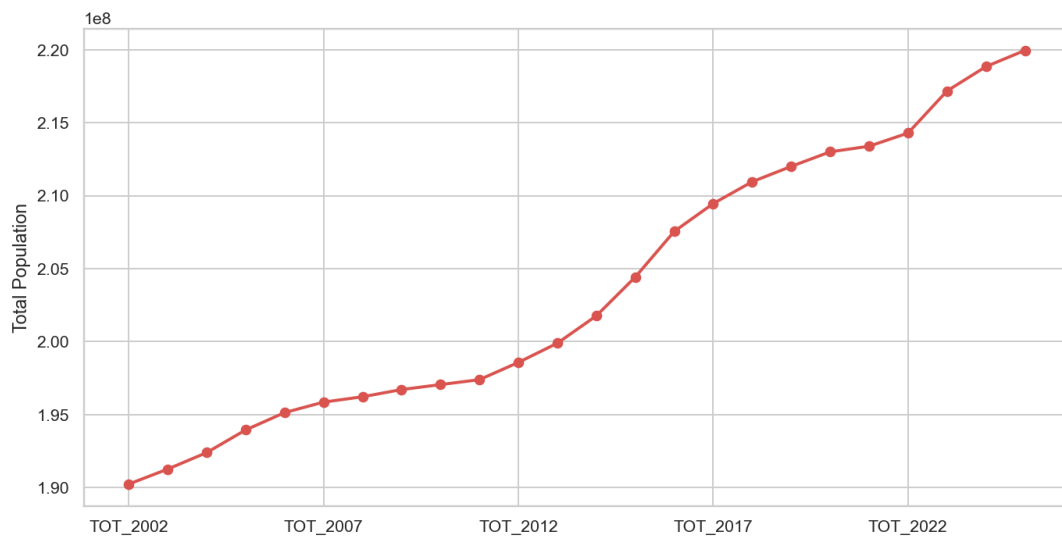
2.5 WP5: Discussion and Results

Objectives and Tasks

In terms of the fifth work package of the project, the goal was to effectively communicate and understand the analytical results. Clear visualization for the analysis output is a key point of this work package. This is especially so in the context of flooding and evacuation, and more so for the key stakeholders of the projects, such as children, elderly and emergency responders. The aim of a WP5 is to take spatial analysis and transform it into an understandable way of communication that can be interpreted by anyone.



Temporal Trend of Population in HQ30 Flood Zones (2002-2025)



• Figure 10: Result

Long-term Population Growth and Exposure Acceleration (2002–2025)

We see that the population is increasing, when taking a long-term look at the demographic trend in Austria. Statistics from Statistik Austria show that it's expected to be around 9,197,213 by 2025, with a 0.54% annual growth rate since 2023, largely thanks to immigration and people settling in cities. When superimposed with hydrological hazard maps, the picture isn't pretty. The population residing in zones that are susceptible to flooding, specifically at the HQ30 recurrence interval, is shooting upwards. The temporal trend graph reveals that after 2017, the number of people living in these zones crossed the 2.15 million mark.

Spatial Polarization of Population Change and Internationalization

When looking at the spatial distribution of the Austrian population from 2002 to 2025, you'll notice a stark geographical divide, with "hotspots" of high growth, largely in the Vienna basin and the western parts of Tyrol and Vorarlberg, where the growth rate regularly tops 42%. Well-known internationalization indices show that in these rapidly growing cities, non-Austrian nationals make up anywhere from 36 to 69 percent of the population, which has the potential to pose challenges to the emergency services. Since the population is expected to swell to 110 people per square kilometer in these areas, the communication strategies for warnings and emergency services have to consider this diversity of languages and the fairly regular turnover of residents.

Evolution of Household Structures and Social Vulnerability (SVI)



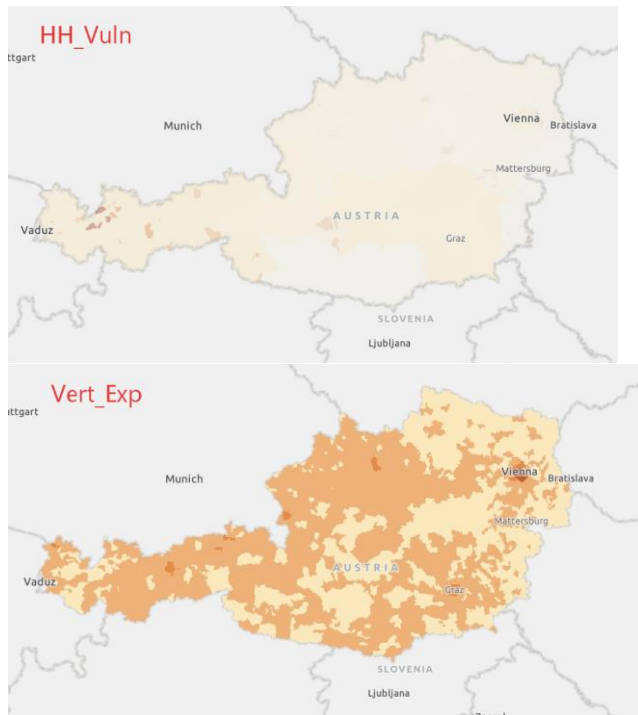
• *Figure 11: Result of SVI*

The spatial visualization of the 2025 structural data reveals a profound systemic shift in flood resilience across Austria, primarily driven by the reduction of average household sizes to 2.17 and the surge of single-person households to 39%. By overlaying the Macro SVI and Local SVI maps, we can contextualize how these demographic shifts materialize as geographic risk clusters. At the national scale, the Macro SVI results position the southern province of Vienna as a clear vulnerability hotspot, as SVI_Score values are above 18.48. As reflected in the indicator system rationale, this peak score is the direct result of the weighted convergence of a rapidly aging population ($OLD_PCT * 0.4$) and a high density of foreign nationals ($FOR_PCT * 0.3$), both facing significant barriers when attempting a rapid, unassisted evacuation. The "scorched" red zones in the Vienna map represent a crucial socio-spatial paradox: although the urban core has the highest infrastructure within the country, the high density of isolated, single-occupancy households forms a type of "vulnerability trap" wherein social capital is thinly spread.

If we now zoom in to review more details for Local SVI maps, it is evident that a high degree of fragmentation is occurring, which was previously glossed over at a macro-level assessment. As an example, it is evident that in our Local SVI map, in border regions, a disproportionately high local-level vulnerability is occurring due to a 20.2% increase in the 65+ demographic profile, in conjunction with a stagnant fertility rate. Unlike the urban region, which emphasized a caregiving deficit or a lack of a younger, more mobile populace, in border regions, there truly is a functional link, as indicated in our ArcGIS Pro model, a high socially weighted isolation index (I_swi). For regions in which a somewhat moderate physical distance (HubDist) from shelters is evident, a caregiving deficit directly contributes to a strikingly high isolation index, which is then further compounded by a flood-evacuation delay index (L_delay). The data strongly supports that, in fact, it appears as though in border regions, a type of 'preparation time,' weighted by the factor of HH_Vuln , effectively counteracts any true lead time that would

presumably exist, so that socially, flood victims in such regions are sequestered from floodwaters, prior to their peak.

Empirical Analysis of Socio-Demographic Drivers and Spatial Vulnerability



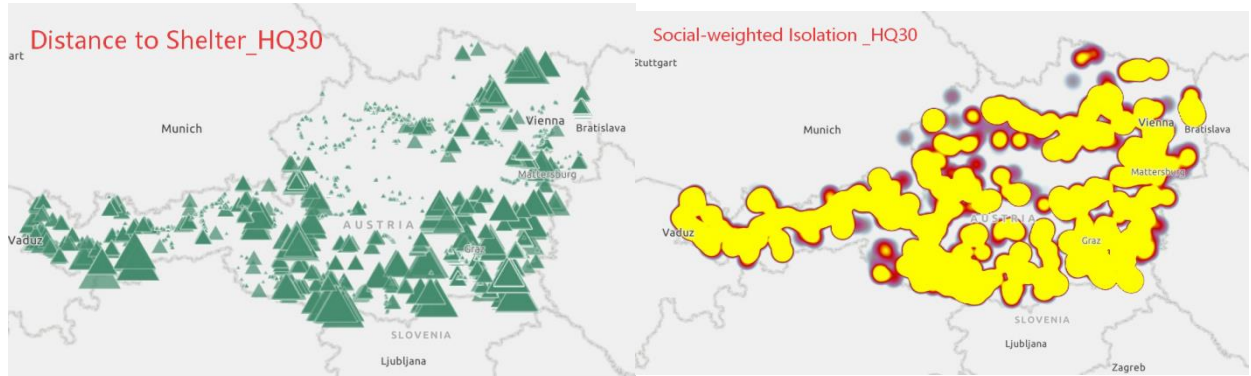
• *Figure 12: Result of Vulnerability*

This 'scorched' risk profile is statistically validated through a combination of high single-occupancy density in high-density populations, coupled with a rapidly aging demographic profile, as epitomized through the HH_Vuln (Household Vulnerability) coefficient. Among these high-scoring 'urban' regions, the presence of dispersed populations inclined to be 'alone' both amplifies the risk profile associated with "emergency entrapment."

This spatial reality can be further understood through the Vert_Exp (Vertical Exposure) index, which points out how the concentration of "aging populations" within these structures has increased the "exposure depth" of the population. Here, Vienna is characterized as a "vulnerability trap" according to its urban isolation. However, through the use of the Local SVI maps, a new type of risk emerges along the border municipalities of the southeastern region. Here, the connection of "vulnerability" is categorized according to a "thinning" of the caregiver-to-dependent ratio, where a 20.2% increase emerges as a function of declining fertility rates among the 65+ demographic. In such a manner, the connection of "vulnerability" can be understood through the increased Socially Weighted Isolation Index (I_{swi}), as revealed through the evidence evident within the (layer). These border municipalities can be understood as functioning as evacuation bottlenecks according to a high HH_Vuln and Vert_Exp\$ measures effectively negate the available lead time that is provided by the early warning systems.

The results thus illustrate that the physical distance to the shelters, HubDist, is not an indicator of safety; rather, the combination of household and demographic exposure creates a "hidden" level of risk wherein these citizens are sequestered socially before ever being affected by the physical floodwaters of the HQ100 expansion in full force.

Comparative Analysis of Physical Distance and Socially Weighted Isolation



• *Figure 13: Result of Distance and Socially Weighted Isolation*

One key observation in judging the results is to recognize the wide discrepancy found in the results from both methods. On the one hand, while the spatial results generated by the HubDist (Euclidean Distance to Shelters model) offer relevant results from the geometric accessibility perspective, the combined index value generated by using $I_{swi} = HubDist * (1 + Local_SVI)$ indicates that there is a hidden "layer of socially amplified isolation" not accounted for by the purely geometric results. From the results obtained at the macro level, grouped under Vienna, areas where there is a high value in SVI, greater than 18.48, it can be noted that while areas may have moderate HubDist, there is a huge scale factor involved with respect to the results obtained from using I_{swi} , given that household vulnerability is high (HH_Vuln) and that there is a presence of 39% population in single-person households.

Additionally, the local-scale results along the southeastern border municipalities reveal intriguing results, presenting isolating "hotspots" where the physical accessibility to emergency centers is somehow cancelled out by socio-economic impediments. In these "hotspots," the combination between high values of I_{swi} and Vert_Exp signifies a state of entrapment. Indeed, while a safe center is physically accessible within one's usual walking range, socio-demographic hindrances such as those imposed by the 20.2% increase in the 65+ demographic subgroup force L_delay (Evacuation Delay) values to surpass the critical time necessary to respond to an HQ100 event, proving conclusively that flood risk management must transcend its topographic character and move toward a socially informed model, whereby it becomes evident that the maximum values recorded by I_{swi} are, in fact, indicative of the real "bottlenecks" in terms of accessibility, owing to limited social capital.

Functionality

Setting up the flood hazard dashboard we had to get to the point of what we wanted it to do. Well-known as a tool that lets users explore the potential impact of floods and their effects on vulnerable groups, this dashboard is designed to be intuitive and interactive. We wanted to make sure that users can compare types of floods using HQ30 and HQ100 layers and are able to understand them in terms of risks it provides to all stakeholders. We needed to map the distribution of people who are at risk, focusing on elderly and children, and check how easily these people can get to emergency shelters with route and travel time indicators as well as path cost function results.

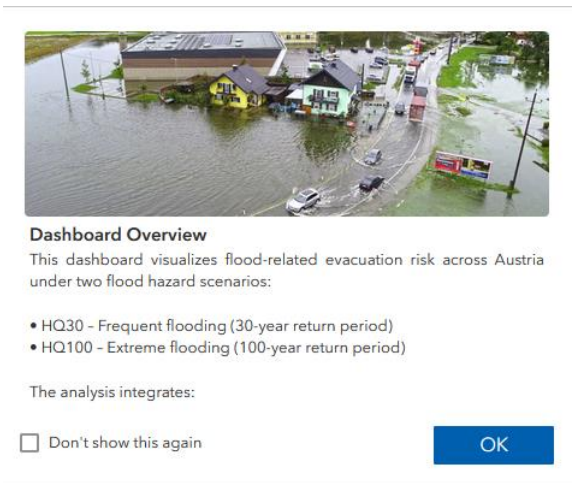
To make the analysis more transparent, the dashboard also includes summary statistics and charts that change as the user goes through the different steps. It is a visual way to help you see how the risks vary from one area to another. The interactivity is intended to get people thinking

about possible disaster and provides a tool to prepare themselves to it, rather than making operational decisions on spot.

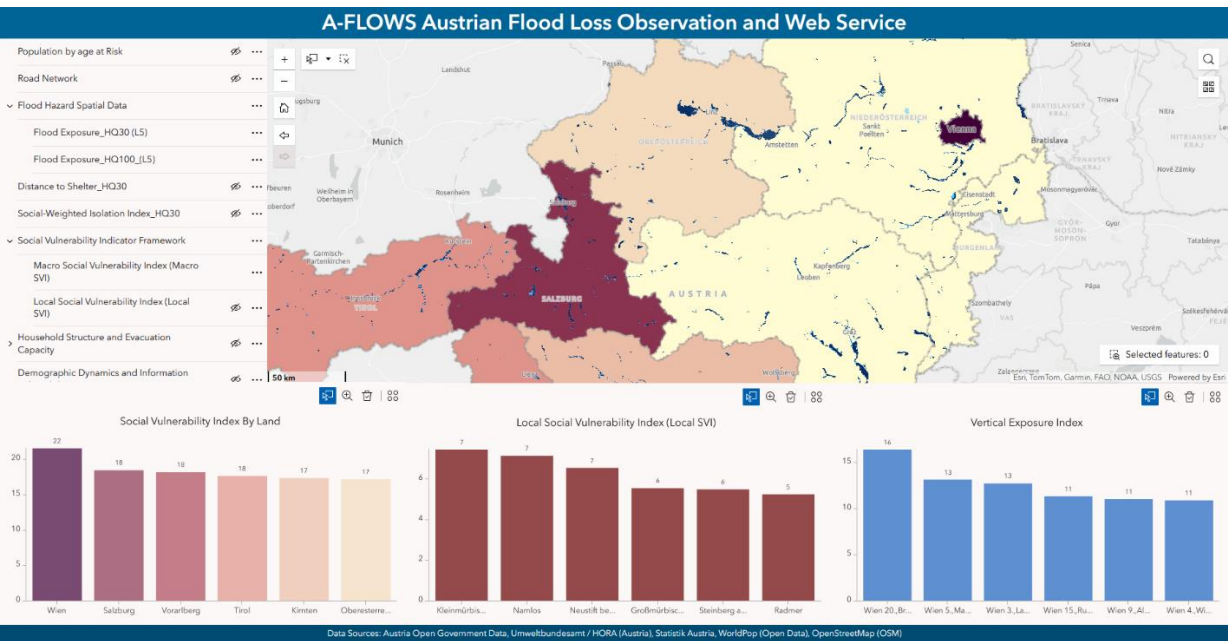
Design

Designing the dashboard in Experience Builder, we wanted to make it so that everyone could understand the data it presents, no matter who they are. Common principles of clear visual language, simple color schemes for high-risk areas, people in harm's way, and accessibility features have been used to create a layout that is intuitive and straightforward.

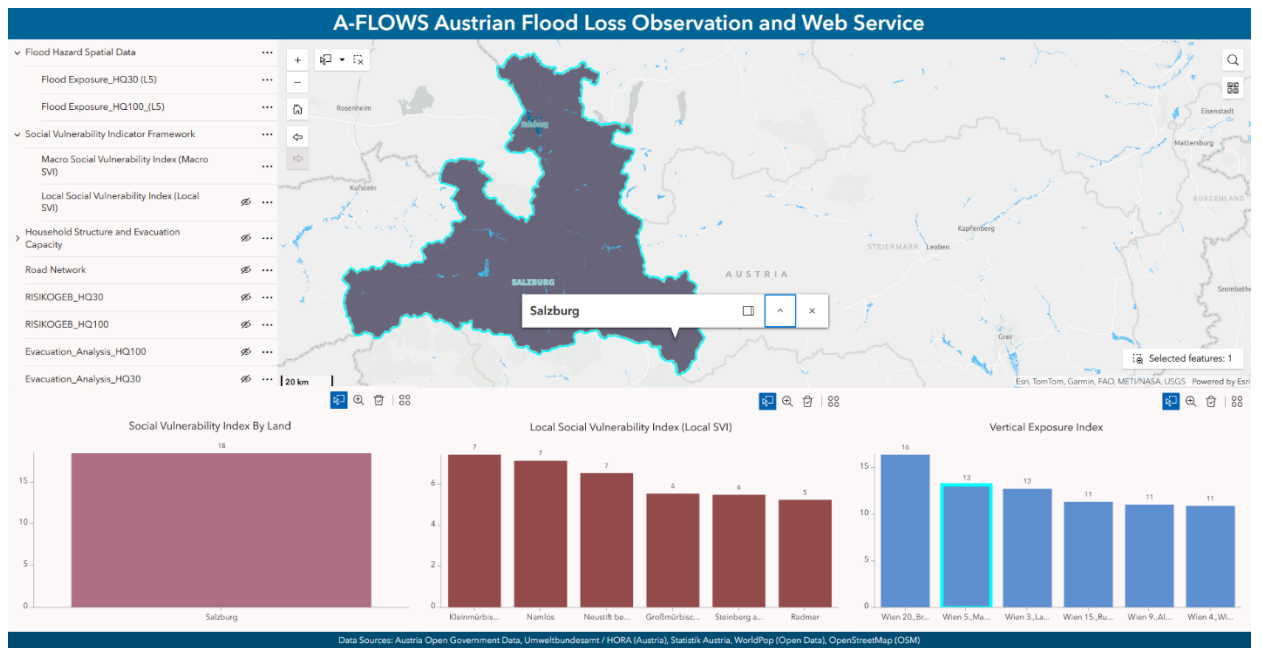
Coming from the guidelines of cartographic visualization and map-centric dashboard design, we have managed to create a tool that presents intricate spatial analysis results in a clear format, making it possible for anyone to quickly get the picture of the key points and connections, without needing a background in GIS. Since we prepared a big number of analyzed layers, we decided to build a dashboard around the map showing only the most important indexes in connection to the map.



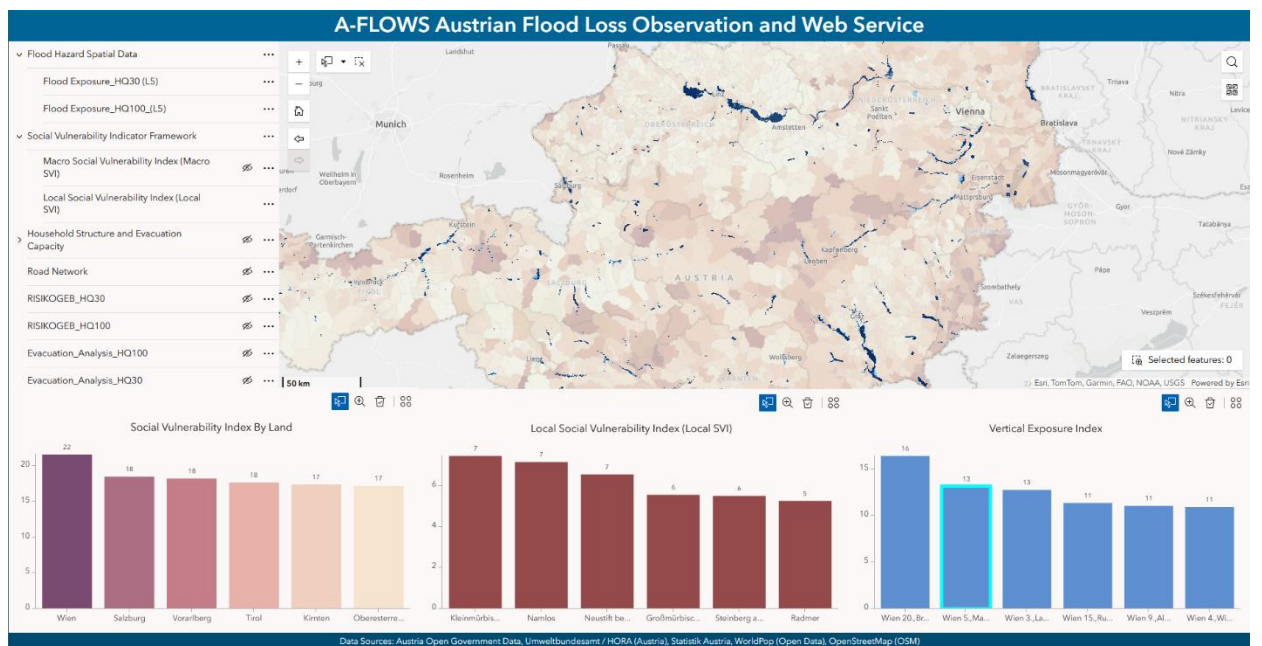
• *Figure 14: Dashboard Overview start window*



• *Figure 15: Macro Social Vulnerability Index*



• *Figure 16: Macro Social Vulnerability Index for Salzburg*



• *Figure 17: Local Social Vulnerability Index*

3. Milestones and Deliverables

Final chapter of the report is dedicated to challenges we faced as a group working on this project and the learnings we achieved during the work. As university students we found that the process itself regardless of how challenging it was will most definitely help us in our future projects and work experiences.

Project Management

Even though A-FLOWS is a relatively small project created as a part of the university course, we faced numerous different problems in different aspects of work. One of the biggest problems for our group were time management of a project. This fact highlights the importance of WP1: realistic planning and continuous coordination of a project team are crucial for it's successful completion.

Table 2: Milestones of project A-FLOWS

Name		Date Completion
M1	Project Management	01.11.2025
M2	Literature Review	01.12.2025
M3	Data Collection	20.12.2025
M4	Methods	27.01.2026
M5	Discussion	31.01.2026

During the work on this project, we encountered several problems that affected the dates we finished few of the milestones. Most of them were caused by complexity of data collected and caused the need of more time to make the analysis and visualization.

Table 3: Deliverables and GitLab connections of project A-FLOWS

Name	Link
D1 Project overview document	https://git.sbg.ac.at/s1104079/Austrian_Flood_Loss_Observation_and_Web_Service/-/blob/main/Project%20Management/Project_Overview_version2.docx?ref_type=heads
D2 GitLab documentation	https://git.sbg.ac.at/s1104079/Austrian_Flood_Loss_Observation_and_Web_Service/-/wikis/Project--Overview?redirected_from=home
D3 Final project report	Final report document (A-FLOWS_Final_Project.docx)
D4 Spatial database (PostGIS)	PostgreSQL/PostGIS database containing flood hazard zones (HQ30, HQ100), demographic indicators, vulnerability indices, and evacuation analysis results
D5 Published Web Services	WMS/WFS services published via GeoServer: • Flood hazard zones (HQ30, HQ100) • Zählsprenkel-based exposure and vulnerability layers • Evacuation accessibility and routing outputs
D6 Published metadata catalogue	https://git.sbg.ac.at/s1104079/Austrian_Flood_Loss_Observation_and_Web_Service/-/tree/main/metadata?ref_type=heads
D7 Interactive dashboard	ArcGIS Experience Builder dashboard for interactive exploration of flood exposure, vulnerable populations, and shelter accessibility

Spatial Data Infrastructure

We gained a lot of practical experience with working within a Spatial Data Infrastructure framework, when developing the A-FLOWS project. Coming from theoretical knowledge of SDI concepts and standards, the project gave us a chance to put these ideas into practice. However, we encountered some issues when merging different datasets, dealing with different spatial reference systems and getting the analysis results into a format that's ready for standardized storage and visualization. A centralized geodatabase and sending the data to interoperable geospatial services really made us see how all these SDI components, like data hosting, metadata, and service publishing, work together in the real world.